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Temperature dependence of the thermo-optic coefficient of lithium niobate, from 300 to 515 K in the visible and infrared regions

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The temperature dependence of the thermo-optic coefficients of lithium niobate has been measured in the temperature range between 320 and 515 K for both wavelengths of 632 and 1523 nm and for ordinary and extraordinary refractive indices. The experimental data, carried out by means of a simple interferometric technique, have been compared with two theoretical models that follow different approaches for treating the thermo-optic effect. The experimental results show good agreement with the theoretical ones. © *2005 American Institute of Physics*.
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Lithium niobate (LN) is one of the most used materials in optoelectronics. It presents a very interesting combination of properties and characteristics that makes possible the realization of different classes of devices. Such optical, acoustical, piezoelectrical, and photorefractive properties have been widely studied and exploited in optoelectronics and photonics.[1] As a matter of fact, LN has been employed in many optical devices, such as waveguides, high-speed modulators,[2] beam deflectors, optical frequency converters, and tunable sources of coherent light for spectroscopic applications.[3]

The knowledge of the refractive-index variation with temperature, that is the thermo-optic coefficient (TOC), is of fundamental importance for the correct design and operation of modern optoelectronic integrated devices and circuits.[4] This is especially true for devices where the wavelength dependence scales with the refractive index, such as in distributed feedback filters, interferometric modulators and switches, wavelength demultiplexers, and, of course, in all thermo-optic-effect-based devices. Unfortunately, while much of the data on the absolute value of the refractive index are reported in literature, in most cases a precise value of the thermo-optic coefficient, and its dependence on the temperature, is incomplete, also at important visible and infrared transmission wavelengths. This is also the case of lithium niobate. Therefore the measurement of the ordinary and extraordinary TOCs in LN and the characterization of the thermo-optic effect with the temperature are still important areas of further studies. Until now, only a few formulations of the temperature dispersion of the LN refractive indices can be found in literature. Ghosh[4] gives a theoretical analysis of the TOC and its dependence on the wavelength and temperature. This model is based on the thermal-expansion coefficient, the temperature coefficient of excitonic band gap, and an isentropic band gap. Basic experimental and theoretical works on the optical properties of LN have been done by Schlarb and Betzler in Ref. 5. The authors derived a generalized Sellmeier equation that takes into account all the effects and the parameters influencing the refractive-index value, such as the wavelength, the composition, and the temperature. However, temperature stability and tunability, calculated by several authors from the different measured LN thermo-optic coefficients, often widely differ from those of experiments.[6,7]

In this Letter we propose and discuss a direct measurement of the TOC for LN in the temperature range between 300 and 515 K, at the wavelengths of 632 and 1523 nm, for both ordinary and extraordinary crystal refractive indices. Many different techniques have been used to measure TOC, most of which necessitate the manufacture of a sample or a structure explicitly required for the measure itself, such as those based on prism-shaped specimens or on diffraction gratings. A simpler technique, applicable mainly to semiconductor materials, was presented[8] for room temperature, and successively improved,[9] extending the temperature range up to 550 K. This technique is based on the multiple-beam interference principle that arises in a Fabry-Perot cavity interrogated by a monochromatic light beam. The intensity of the radiation transmitted by the cavity shows a periodic fringe pattern when measured as a function of the sample temperature. The temperature difference between two subsequent

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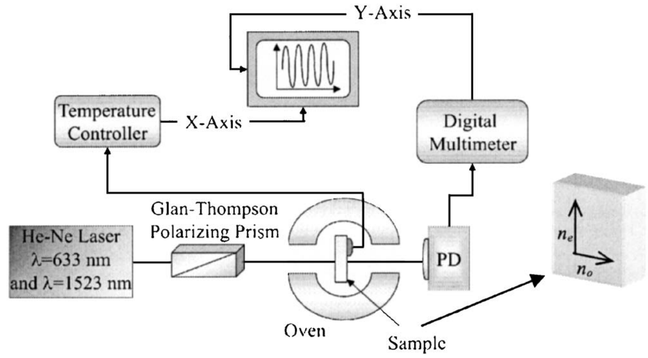

FIG. 1. Experimental setup used to measure the thermo-optic coefficients. An optical scheme of the lithium niobate sample is also reported.

transmission fringe maxima or minima allows the direct determination of the TOC (Ref. 8) from the expression:

$$\frac{dn_i}{dT} = \frac{\lambda_L}{4l\Delta T_{\pi/2}} n_i \alpha(T), \quad i = o,e, \tag{1}$$

where λ_L is the wavelength of probe light, l is the etalon length, n_i is the ordinary (o) or extraordinary (e) refractive index, $\alpha(T)$ is the thermal expansion of the material in the direction of light propagation, and $\Delta T_{\pi/2}$ is the temperature variation required to cause a complete detuning of the optical cavity. In order to measure the TOC for ordinary and extraordinary refractive indices we have utilized an x-cut single domain crystalline LN etalon (by Crystal Technology Inc.), obtained by optical polishing the two parallel crystal facets; the cavity thickness is 470 μm. The thermal expansion, in the x direction, is given by[10]

$$\alpha(T) = 1.44 \times 10^{-5} + 7.1 \times 10^{-9}(T - 298)\ (\mathrm{K}^{-1}). \tag{2}$$

A simple schematic of the experimental setup is reported in Fig. 1. In order to take into account the LN crystal birefringence, it is mandatory to perform the transmission fringe acquisition controlling the probe light beam polarization. In other words, it is important to evaluate the transmission fringe pattern through the etalon with linearly polarized probe beams, in order to investigate both the ordinary and extraordinary refractive indices.

The experimental results for the wavelengths of 632 and 1523 nm are reported in Figs. 2 and 3, respectively. The experimental uncertainties on the data are about 10^{-7} and 10^{-6} K^{-1} for the TOCs relative to the ordinary and extraordinary refractive indices, respectively. In the given temperature range and at the wavelength of 632 nm, a best fitting procedure returns the following values (solid line in Fig. 2):

$$\frac{dn_o}{dT} = -1.7 + 6.9 \times 10^{-3} T\ (10^{-5}\ \mathrm{K}^{-1}),$$

$$\frac{dn_e}{dT} = -2.6 + 22.4 \times 10^{-3} T\ (10^{-5}\ \mathrm{K}^{-1}). \tag{3}$$

For the TOC at $\lambda_L = 1523$ nm, we obtain the following results (solid line in Fig. 3):

$$\frac{dn_o}{dT} = -0.9 + 3.0 \times 10^{-3} T\ (10^{-5}\ \mathrm{K}^{-1}),$$

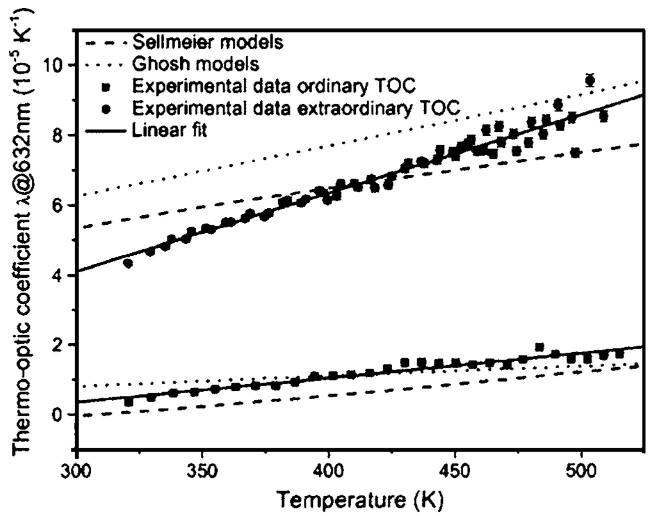

FIG. 2. Measured and theoretical values of ordinary (■) and extraordinary (♦) thermo-optic coefficients at 632 nm. Solid lines: linear fit. Dash lines: Sellmeier-based model. Dot lines: Ghosh model.

$$\frac{dn_e}{dT} = -2.6 + 19.8 \times 10^{-3} T\ (10^{-5}\ \mathrm{K}^{-1}). \tag{4}$$

It is interesting to note that the extraordinary refractive index has a greater value of TOC, and its temperature dependence is quite stronger with respect to the TOC of the ordinary refractive index.

For comparison purposes, we have considered the two different theoretical models previously cited. The Ghosh model takes into account the thermal-expansion coefficient, the temperature coefficient of the excitonic band gap, and the isentropic band gap that lie in the UV region. In this approach, the TOC is given by the following expression:[4]

$$2n_i \frac{dn_i}{dT} = G_i \frac{\lambda^2}{(\lambda^2 - \lambda_{\mathrm{ig}}^2)} + H_i(T) \left[\frac{\lambda^2}{(\lambda^2 - \lambda_{\mathrm{ig}}^2)} \right]^2, \quad i = o,e, \tag{5}$$

where λ is the wavelength of incident light and λ_{ig} is the wavelength of isentropic band gap lying in the UV region.

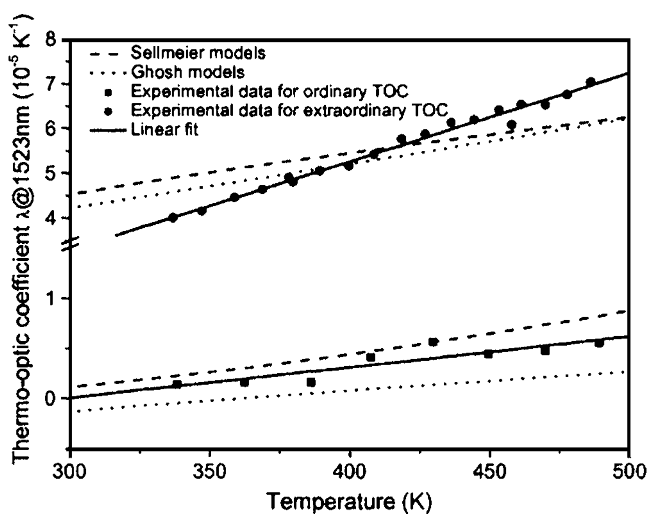

FIG. 3. Measured and theoretical values of ordinary (■) and extraordinary (♦) thermo-optic coefficients at 1523 nm. Solid lines: linear fit. Dash lines: Sellmeier-based model. Dot lines: Ghosh model.

The constants G_i and H_i are related to the thermal-expansion coefficient α and to the temperature coefficient of the excitonic band gap E_{eg}, respectively. They are given by the relations $G_i = -3\alpha K_i^2$ and $H_i = -(1/E_{eg})(dE_{eg}/dT)$. The constant K_i is related to the less dispersive refractive index $n_{i,\infty}$ in the IR region, and is given by $K_i^2 = n_{i,\infty}{}^2 - 1$. In Ref. 11, this model has been applied to the determination of the TOC of several nonlinear crystals, among which is LN; it is also reported the temperature dependence of the optical parameters H_i.

The second model that we have taken into account is that by Schlarb and Betzler.[5] In their work, on the basis of the generalized Sellmeier equation, all refractive-index-dependent effects in LN have been calculated in the VIS and IR spectra, in the temperature range of 50–600 K. Without entering in the details of the model, we underline that a straightforward numerical estimation of TOC has been carried out adopting the fitted parameters reported in Ref. 5.

The theoretical temperature dependence of the TOC calculated according to these models is reported in Figs. 2 and 3, for the red and infrared radiations, respectively. A good agreement between the two models and our experimental results on the TOC of the ordinary refractive index is obtained. In fact, the experimental values are included between the Ghosh- and the Sellmeier-based models. For the TOC relative to the extraordinary refractive index, the experimental results give stronger temperature dependence with respect to the theoretical models.

In conclusion, we have adopted a simple technique to characterize the ordinary and extraordinary thermo-optic coefficients of LN at the wavelength of 632 and 1523 nm, in the temperature range of 300–515 K. For both wavelengths, the extraordinary index TOC shows a value which is about two times the ordinary index TOC. Moreover, the first one shows a stronger dependence from temperature. The experimental data have been compared with the Ghosh- and a generalized Sellmeier-based models, showing good agreement with both of them.

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